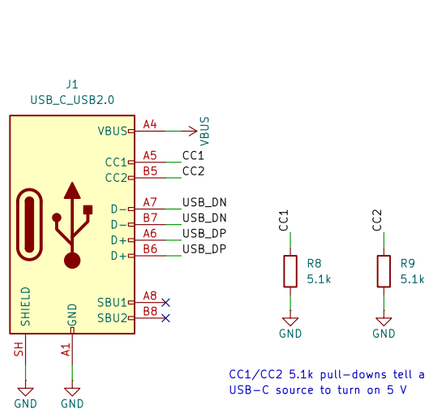
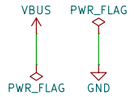
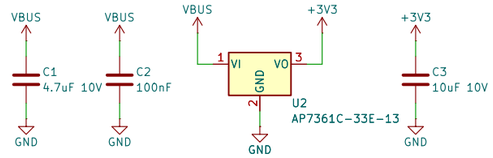
USB-C: 5 V input and programming

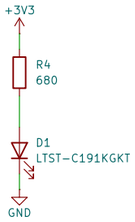




CC1/CC2 5.1k pull-downs tell a USB-C source to turn on 5 V

3.3 V regulator and power LED

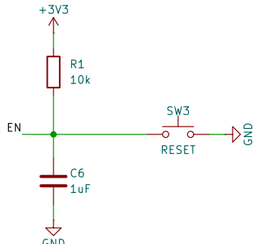




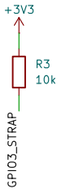
In: 4.7uF + 100nF Out: 10uF (>= 2.2uF ceramic per datasheet)

LED: (3.3 V − 2.0 V) / 680 = 1.9 mA

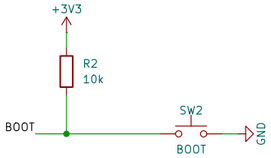
Reset, boot and strapping



EN delay: 10k x 1uF = 10 ms

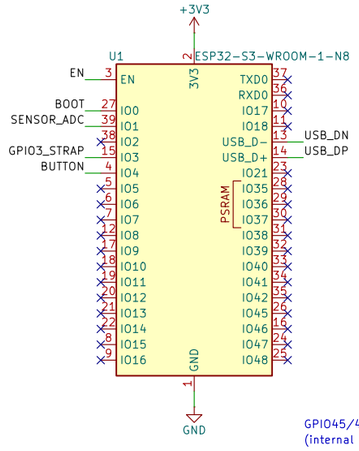


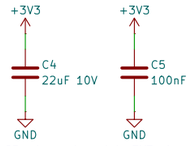
GPIO3 has no internal pull; datasheet says it must not float



Hold BOOT, tap RESET = download mode

ESP32-S3-WROOM-1 module

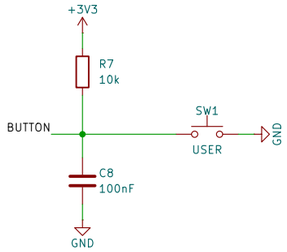




C4, C5: place at module 3V3 pin

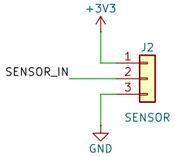
GPIO45/46: leave open (internal pull-downs)

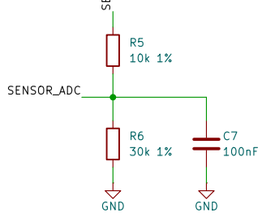
User button (active low)



Debounce: 10k x 100nF = 1 ms

Analog sensor input (0 to 3.3 V)





J2 pin 1: 3V3, pin 2: SIG, pin 3: GND

3.3 V x 30k / 40k = 2.475 V (ADC max 3.1 V)
RC corner: 1 / (2 pi x 7.5k x 100nF) = 212 Hz

Mounting holes (M3)



ESP32-S3-WROOM-1, USB-C 5 V in, AP7361C 3.3 V LDO

Sea Otter 2027 bike build, EE subteam

Sheet: /

File: kicad bike project.kicad_sch

Title: Basic Sensor Interface PCB

Size: A3	Date: 2026-10-09	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/1